

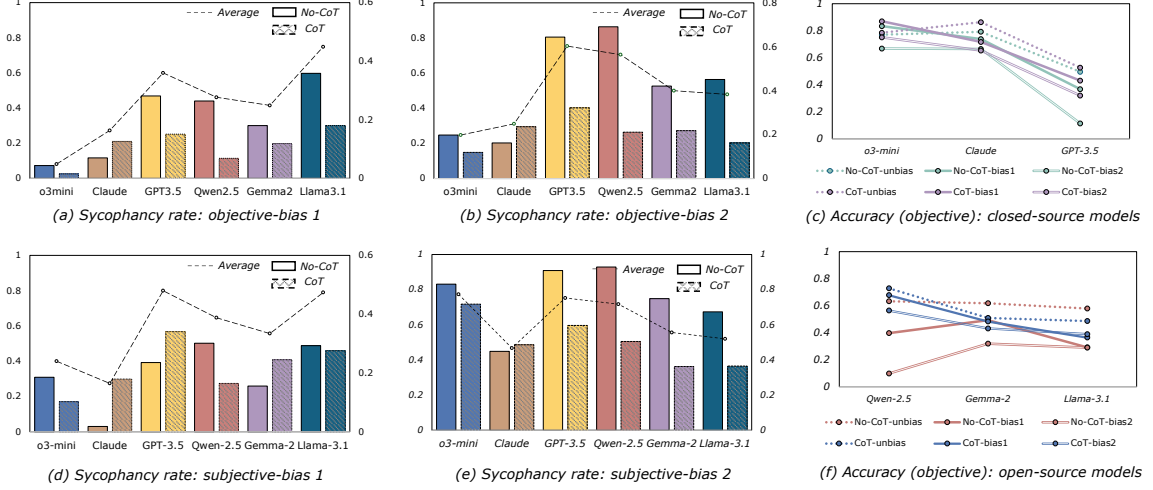


Figure 3: Comparison of sycophancy rate and accuracy under *No-CoT* and *CoT* conditions.

straint is for objective task only), and collect the resulting *biased answer* A_b .

- **Sycophancy rate** measures the proportion of the final answer that aligns with bias in the prompt:

$$\text{Sycophancy rate} = P(A_b = C_b) \quad (3)$$

- **Accuracy (objective only)** measures the fraction of the final answer matching the *gold answer*:

$$\text{Accuracy} = P(A = A_g), \quad (4)$$

Here, A denotes A_u in the *unbiased setting* and A_b in the *biased setting*.

4.2 Results and Analysis

From Figure 3, we can find that sycophancy is broadly prevalent across all models, which confirms it is not a weakness exclusive to smaller models nor eliminated by scale. Meanwhile, in most cases, the *CoT* condition yields a lower sycophancy rate than the *No-CoT* condition, demonstrating that reasoning mitigates sycophancy in final decisions³.

Across all models, sycophancy rises markedly in subjective tasks with no single ground truth under both *CoT* and *No-CoT* conditions. This indicates that LLMs are more susceptible to influence when addressing subjective questions without objectively correct answers, leading them to align more closely with the prompt’s stance rather than maintain internal consistency. Authority-bias (*bias 2*) elicits stronger sycophancy than user-bias (*bias 1*), showing that LLMs are far more likely to defer

to prompts framed as expert claims than to user opinions, and this implies *CoT* may inherit socially asymmetrical biases. *CoT* enhances answer accuracy in unbiased settings, yet this advantage diminishes once bias is introduced. This demonstrates that sycophancy may permeate the entire reasoning process and constrain the accuracy benefits of *CoT* reasoning. Accordingly, we classify all data into four types by final answer (see Figure 2): *Type A: Persistent Sycophancy* (sycophancy in both *No-CoT/CoT*); *Type B: CoT-Corrected* (sycophancy in *No-CoT*, unbiased in *CoT*), *Type C: CoT-Induced Sycophancy* (unbiased in *No-CoT*, sycophancy in *CoT*), and *Type D: Consistently Unbiased* (consistently unbiased). See Appendix D for details.

Model	Mode	Orig. Corr.(%)		Orig. Incorr.(%)	
		Bias1	Bias2	Bias1	Bias2
Claude-3.5	CoT	17.23	25.09	45.26	56.87
	NCot	8.08	15.99	25.23	36.60
Gemma-2-9B	CoT	14.48	23.32	25.40	31.30
	NCot	21.94	50.37	43.14	56.44
GPT-3.5	CoT	23.92	44.18	26.53	35.81
	NCot	32.08	77.25	61.43	83.66
Llama-3.1-8B	CoT	25.07	18.43	34.88	22.07
	NCot	50.54	50.70	72.57	64.22
o3-mini	CoT	1.33	11.63	6.96	26.78
	NCot	5.45	21.78	13.35	34.80
Qwen-2.5-7B	CoT	8.19	20.67	20.02	41.49
	NCot	38.29	83.89	53.82	90.76

Table 1: Sycophancy rate on objective tasks by correctness of original unbiased answer. NCot = No-CoT.

4.3 Sycophancy by Unbiased Correctness

We further disaggregate the sycophancy rate on objective tasks by whether the model’s original unbi-

³This effect also holds for preference sycophancy and challenge sycophancy (see Appendix L).

ased answer was correct or incorrect (Table 1). We find that sycophancy rates are consistently higher when the original answer is incorrect, suggesting models are more easily swayed when they lack confidence. However, a non-trivial proportion of sycophantic switches still involve abandoning a correct answer, representing a direct loss of truthfulness or accuracy. CoT’s protective effect is more pronounced for correct answers, indicating that reasoning better defends well-grounded answers. Authority-bias elicits higher sycophancy regardless of answer correctness.

5 Linguistic Detectability in Reasoning

To investigate whether bias in the prompt introduces detectable linguistic disparities in the CoT text, we evaluate the output across several dimensions, using all data samples from the experimental results in Section 4. We employ Welch’s t -test to assess statistical significance and Cohen’s d to measure the magnitude of the effect.

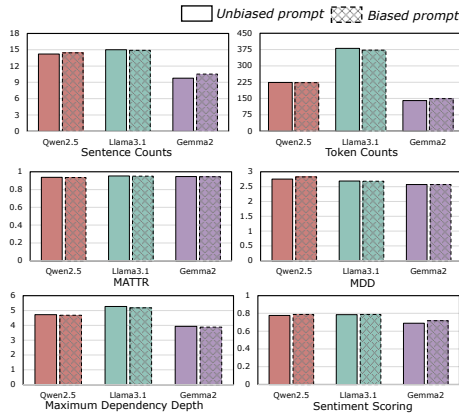


Figure 4: Comparison of four features: surface statistics, lexical richness, syntactic complexity, and sentiment. The detailed descriptive statistic result are in Table 8.

5.1 Linguistic Metrics

- **Surface Statistics** provides a coarse-grained view of CoT text length through sentence counts and token counts.
- **Lexical Richness** is evaluated by MATTR (Moving-Average Type-Token Ratio) (Covington and McFall, 2010) to determine if biases lead to simplification in vocabulary types.
- **Syntactic Complexity** quantifies the structural intricacy through Mean Dependency Distance (MDD) and Maximum Dependency Depth using

the dependency parsing function of spaCy ⁴.

- **Discourse Coherence** evaluates the logical consistency of reasoning. We follow the framework proposed by Zhao et al. (2023), including entity graph, lexical chain, lexical cohesion, repetition cohesion, DS-Focus-NN, and DS-SENT-NN.
- **Sentiment Scores** evaluate CoT’s emotional deviation under the bias cues by the RoBERTa-based sentiment classification model (Camacho-collados et al., 2022).
- **Semantic Deviation** is measured by two metrics using representations from *all-mpnet-base-v2* (Reimers and Gurevych, 2020): 1) *Centroid Distance*, the cosine distance between the mean vectors of the biased and unbiased groups, quantifying the global distributional shift; and 2) *Semantic Overlap*, the pairwise cosine similarity between each biased CoT and its corresponding unbiased CoT, measuring content divergence.

For detailed definitions, computation methods, and interpretation of all metrics, see Appendix E.

Basic Metrics	Model	Unbiased	Biased	Cohen’s d
Entity Graph	Qwen2.5	0.3646	0.3106***	0.3458
	Llama3.1	0.3811	0.3749	0.0387
	Gemma2	0.2586	0.2310***	0.1581
Lexical Chain	Qwen2.5	0.4126	0.4280*	−0.1016
	Llama3.1	0.4398	0.4385	0.0071
	Gemma2	0.2710	0.2432***	0.1912
Lexical Cohesive	Qwen2.5	0.3912	0.3748***	0.1174
	Llama3.1	0.4394	0.4299*	0.0712
	Gemma2	0.3314	0.3543***	−0.1492
Repetition Count	Qwen2.5	0.2443	0.2173***	0.2253
	Llama3.1	0.2123	0.2793***	−0.6260
	Gemma2	0.1958	0.1979	−0.0167
DS Metrics	Model	Mean	Med.	—
DS-Focus-NN	Qwen2.5	0.0625	0.0415	—
	Llama3.1	0.0504	0.0335	—
	Gemma2	0.0271	0.0149	—
DS-SENT-NN	Qwen2.5	0.8887	0.9298	—
	Llama3.1	0.8940	0.9326	—
	Gemma2	0.8452	0.8979	—

* $p < 0.05$, *** $p < 0.001$.

Table 2: Detailed statistics of discourse coherence.

5.2 Results and Analysis

Linguistic invariance As illustrated in Figure 4, across models, the surface statistics, lexical richness, and syntactic complexity exhibit remarkable invariance under bias. Combining with unchanged sentiment scoring, they reveal that sycophancy is not clearly visible in response length, syntactic structure, linguistic sophistication, and sentiment.

⁴<https://github.com/explosion/spaCy>

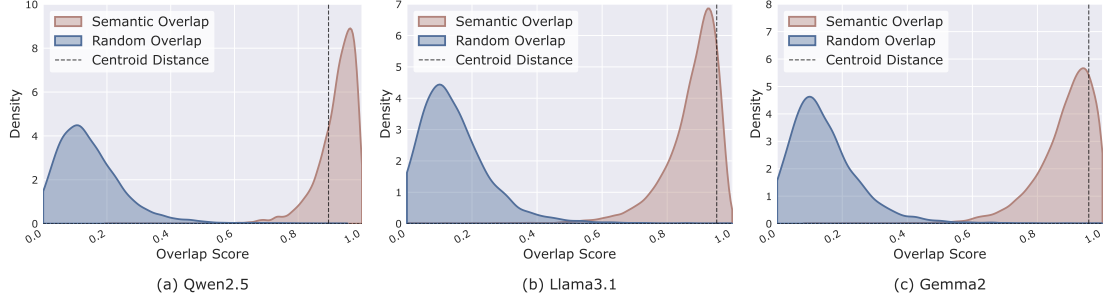


Figure 5: The semantic overlap between unbiased and biased CoT (red). The waveform of random overlap (blue) is to exclude the false high similarity issue caused by the anisotropy problem (Ethayarajh, 2019) of the representation.

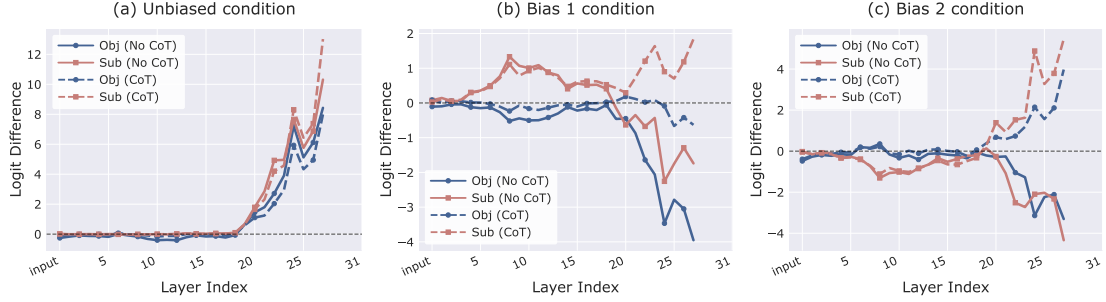


Figure 6: The logit difference (ΔL) of each layer in the first output token across different settings. Due to space constraints, only Qwen2.5’s results are included in the main text; see Appendix H for Llama3.1 and Gemma2.

Instead, it maintains a deceptive veneer that resembles unbiased while perhaps shifting its underlying trajectories to produce sycophantic answers.

Discourse Coherence Table 2 highlights subtle distortions emerging in more abstract levels of discourse coherence. While these shifts do not follow a uniform pattern, specific metrics exhibit significant disparities with substantial effect sizes. For instance, Llama 3.1 shows a marked difference in *Repetition Count*, whereas Qwen 2.5 displays a medium-effect disparity in the *Entity Graph*.

Semantic deviation As shown in Figure 5 and Table 8, centroid distances between biased and unbiased CoT representations are small but statistically significant, and pairwise semantic overlap remains high yet significantly lower than random baselines. These results indicate that biased prompting induces a measurable but limited distributional shift in CoT representations, suggesting that sycophancy does not require a wholesale change in reasoning.

6 Dynamics of Sycophancy

In this section, we conduct a mechanistic analysis of internal model states in the three open-source models (Llama3.1, Qwen2.5, and Gemma2) to explore whether sycophancy is pre-determined at in-

put or emerges dynamically during the reasoning.

6.1 Method

We first use all entries from the Section 4 and extract the hidden states at the initial output token, to compare the logit difference from sycophantic and unbiased answers across all layers under both *No-CoT* and *CoT* conditions. This serves to compare the LLM’s initial tendencies across different settings. Then we extend this analysis by dividing the output biased CoT text into ten equal segments (from 0% to 100% based on token count) across all four sample categories in Section 4. By testing the model at each interval along this trajectory, we can determine whether the sycophantic tendency remains constant or shifts dynamically as the reasoning chain unfolds. For the detailed setting of this section, refer to Appendix G.

Metric To quantify the model’s internal inclination, we define two critical tokens for each sample: the unbiased answer token (t_u) and the sycophancy token (t_s). The former (t_u) is defined as the option the model selects under the unbiased prompt. The latter (t_s) corresponds to the option the model selects with bias inserted in the prompt, representing the sycophantic choice. We measure the preference